Concrete mix proportioning as per IS 10262 : 2019

- ✓ Grade of Concrete: M 25
- ✓ Nominal maximum size of aggregate: 20 mm
- ✓ Type of aggregate: Crushed angular aggregate
- ✓ Workability: slump value of 50 mm
- ✓ Type of exposure: Moderate (for reinforced concrete)
- ✓ Cement: OPC (ordinary Portland cement) 53 Grade
- ✓ Specific gravity of cement : 3.15
- ✓ Specific gravity (SSD) of coarse aggregate : 2.65
- ✓ Specific gravity (SSD) of fine aggregate (sand) : 2.62
- ✓ Fine aggregate: conforming to Grading Zone II, Table 9, IS 383 : 2016

SSD: Saturated and surface-dry

Concrete mix proportioning as per IS 10262 : 2019

Procedure

1. Calculation of target mean compressive strength

i) $F_m = F_{ck} + ks$

ii) $F_m = F_{ck} + X$

where F_m = target mean compressive strength at the age of 28 days

F_{ck} = characteristic compressive strength at the age of 28 days

k = probability factor

s = standard deviation

X = factor based on grade of concrete

IS 10262 : 2019

Table 1 Value of X
(*Clause* 4.2)

Sl No.	Grade of Concrete	Value of X
(1)	(2)	(3)
i)	*M*10, *M*15	5.0
ii)	*M*20, *M*25	5.5
iii)	*M*30, *M*35, *M*40, *M*45, *M*50, *M*55, *M*60	6.5
iv)	M65 and above	8.0

IS 10262 : 2019

Table 2 Assumed Standard Deviation
(*Clause* 4.2.1.3)

Sl No.	Grade of Concrete	Assumed Standard Deviation N/mm^2
(1)	(2)	(3)
i)	*M*10, *M*15	3.5
ii)	*M*20, *M*25	4.0
iii)	M30, M35, M40, M45, M50, M55, M60	5.0
iv)	*M*65, *M*70, *M*75, *M*80	6.0

NOTES

1 The above values correspond to good degree of site control having proper storage of cement; weigh batching of all materials; controlled addition of water; regular checking of all materials; aggregate grading and moisture content; and regular checking of workability and strength. Where there are deviations from the above, the site control shall be designated as fair and the values given in the above table shall be increased by 1 N/mm^2.

2 For grades M65 and above, the standard deviation may also be established by actual trials based on assumed proportions, before finalizing the mix.

Concrete mix proportioning as per IS 10262 : 2019

✓ Generally, the standard deviation is estimated from a number of available test strength of samples (minimum 30).

✓ When sufficient test results for a particular grade of concrete are not available, the value of standard deviation may be assumed from Table 2 (IS 10262 : 2019) for the proportioning of mix in the first instance.

i) $F_m = 25 + 1.65 \times 4 = 31.60$ N/mm^2

ii) $F_m = 25 + 5.5 = 30.50$ N/mm^2

The higher value is selected and thus, the target mean compressive strength is 31.60 N/mm^2.

Concrete mix proportioning as per IS 10262 : 2019

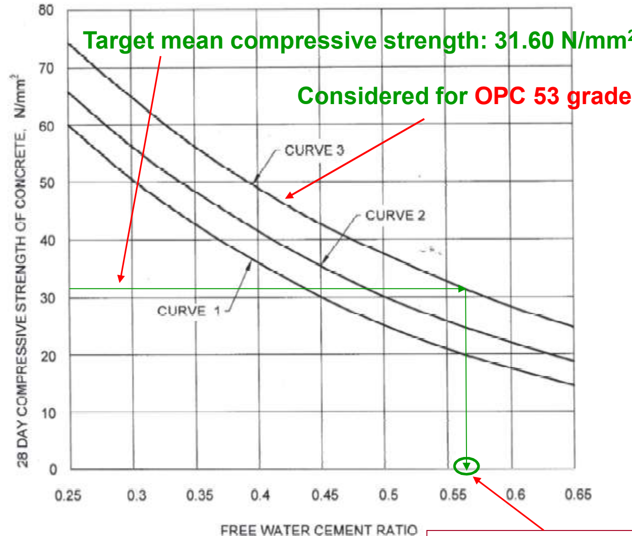
2. Selection of w/c ratio (water-to-cement ratio)

i) Free w/c ratio = 0.56 (From Fig. 1, IS 10262 : 2019)

ii) Maximum free w/c ratio = 0.50 (From Table 5, IS 456 : 2000)

✓ Select lower of the two w/c ratio values. Thus, w/c ratio = 0.50.

IS 10262 : 2019



Free w/c ratio: 0.56

Curve 1 : for expected 28 days compressive strength of 33 and < 43 N/mm².
Curve 2 : for expected 28 days compressive strength of 43 and < 53 N/mm².
Curve 3 : for expected 28 days compressive strength of 53 N/mm² and above.

NOTES

1 In the absence of data on actual 28 days compressive strength of cement, the curves 1, 2 and 3 may be used for OPC 33, OPC 43 and OPC 53, respectively.

2 While using PPC/PSC, the appropriate curve as per the actual strength may be utilized. In the absence of the actual 28 days compressive strength data, curve 2 may be utilized.

Fig 1. Relationship Between Free Water Cement Ratio and 28 Days Compressive Strengths of Concrete for Cements of Various Expected 28 Days Compressive Strengths

IS 456 : 2000 (Reaffirmed 2021)

Table 5 Minimum Cement Content, Maximum Water-Cement Ratio and Minimum Grade of Concrete for Different Exposures with Normal Weight Aggregates of 20 mm Nominal Maximum Size

(Clauses 6.1.2, 8.2.4.1 *and* 9.1.2)

Sl No.	Exposure	Plain Concrete			Reinforced Concrete		
		Minimum Cement Content kg/m³	Maximum Free Water-Cement Ratio	Minimum Grade of Concrete	Minimum Cement Content kg/m³	Maximum Free Water-Cement Ratio	Minimum Grade of Concrete
1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
i)	Mild	220	0.60	–	300	0.55	M 20
iii)	Moderate	240	0.60	M 15	300	0.50	M 25
iii)	Severe	250	0.50	M 20	320	0.45	M 30
iv)	Very severe	260	0.45	M 20	340	0.45	M 35
v)	Extreme	280	0.40	M 25	360	0.40	M 40

NOTES

1 Cement content prescribed in this table is irrespective of the grades of cement and it is inclusive of additions mentioned in 5.2. The additions such as fly ash or ground granulated blast furnace slag may be taken into account in the concrete composition with respect to the cement content and water-cement ratio if the suitability is established and as long as the maximum amounts taken into account do not exceed the limit of pozzolona and slag specified in IS 1489 (Part 1) and IS 455 respectively.

2 Minimum grade for plain concrete under mild exposure condition is not specified.

Concrete mix proportioning as per IS 10262 : 2019

3. Entrapped air content

(IS 10262 : 2019)

Table 3 Approximate Air Content
(*Clause* 5.2)

Sl No.	Nominal Maximum Size of Aggregate mm	Entrapped Air, as Percentage of Volume of Concrete
(1)	(2)	(3)
i)	10	1.5
ii)	20	1.0
iii)	40	0.8

For 20 mm nominal maximum size of aggregate, the approximate amount of entrapped air is 1%.

Concrete mix proportioning as per IS 10262 : 2019

4. Water content

For 20 mm nominal maximum size of aggregate, the water content for 50 mm slump is 186 kg.

(IS 10262 : 2019)

Table 4 Water Content per Cubic Metre of Concrete For Nominal Maximum Size of Aggregate
(*Clause* 5.3)

Sl No.	Nominal Maximum Size of Aggregate mm	Water Content[1] kg
(1)	(2)	(3)
i)	10	208
ii)	20	186
iii)	40	165

[1]Water content corresponding to saturated surface dry aggregate.

NOTES

1 These quantities of mixing water are for use in computing cement/cementitious materials content for trial batches.

2 On account of long distances over which concrete needs to be carried from batching plant/RMC plant, the concrete mix is generally designed for a higher slump initially than the slump required at the time of placing. The initial slump value shall depend on the distance of transport and loss of slump with time.

5. Cement content

Cement content = water content / (w/c ratio)

= 186/0.50 = 372 kg/m^3

372 kg/m^3 > 300 kg/m^3 (minimum cement content for 'Moderate' exposure condition, Table 5, IS 456 : 2000)

Concrete mix proportioning as per IS 10262 : 2019

6. Volumetric proportions of coarse aggregate and fine aggregate

(IS 10262 : 2019)

Table 5 Volume of Coarse Aggregate per Unit Volume of Total Aggregate for Different Zones of Fine Aggregate for Water-Cement/Water-Cementitious Materials Ratio of 0.50
(*Clause* 5.5)

Sl No.	Nominal Maximum Size of Aggregate mm	Volume of Coarse Aggregate per Unit Volume of Total Aggregate for Different Zones of Fine Aggregate			
		Zone IV	Zone III	Zone II	Zone I
(1)	(2)	(3)	(4)	(5)	(6)
i)	10	0.54	0.52	0.50	0.48
ii)	20	0.66	0.64	0.62	0.60
iii)	40	0.73	0.72	0.71	0.69

NOTES
1 Volumes are based on aggregates in saturated surface dry condition.
2 These volumes are for crushed (angular) aggregate and suitable adjustments may be made for other shape of aggregate.
3 Suitable adjustments may also be made for fine aggregate from other than natural sources, normally, crushed sand or mixed sand may need lesser fine aggregate content. In that case, the coarse aggregate volume shall be suitably increased.
4 It is recommended that fine aggregate conforming to Grading Zone IV, as per IS 383 shall not be used in reinforced concrete unless tests have been made to ascertain the suitability of proposed mix proportions.

Concrete mix proportioning as per IS 10262 : 2019

Volume of coarse aggregate per unit volume of total aggregate for 20 mm nominal maximum size of aggregate and fine aggregate conforming to Grading Zone II for w/c ratio of 0.50 = 0.62.

Volume of coarse aggregate per unit volume of total aggregate for w/c ratio of 0.50 = 0.62.

Volume of fine aggregate per unit volume of total aggregate = 1 – 0.62 = 0.38

Concrete mix proportioning as per IS 10262 : 2019

7. Quantities of ingredients per unit volume of concrete (1 m^3)

Volume of entrapped air in wet concrete = 0.01 m^3

Mass of cement = 372 kg

$$\text{Volume of cement} = \frac{372}{3.15 \times 1000} = 0.11809 \text{ m}^3$$

Mass of water = 186 kg

Volume of water = 0.186 m^3

Volume of total aggregate = 1 − (0.01 + 0.11809 + 0.186) = 0.68591 m^3

Mass of coarse aggregate = 0.68591 × 0.62 × 2.65 × 1000 = 1126.95 kg

Mass of fine aggregate = 0.68591 × 0.38 × 2.62 × 1000 = 682.892 kg

Concrete mix proportioning as per IS 10262 : 2019

Quantities of ingredients per unit volume of concrete (1 m^3)

Free w/c ratio = 0.50

Water = 186 kg/m^3

Cement = 372 kg/m^3

Fine aggregate (SSD) = 682.892 kg/m^3

Coarse aggregate (SSD) = 1126.95 kg/m^3

Concrete mix proportioning as per ACI Method

- ✓ 28 day target mean compressive strength = 33.5 N/mm^2
- ✓ Concrete: water tight and exposed to fresh water
- ✓ Maximum size of aggregate (MSA) : 25 mm
- ✓ Dry-rodded bulk density of coarse aggregate = 1580 kg/m^3
- ✓ Fineness modulus of sand = 2.60
- ✓ Slump = 80 mm
- ✓ Cement: OPC (ordinary Portland cement)
- ✓ Specific gravity of cement (S_c) = 3.15

Concrete mix proportioning as per ACI Method

- ✓ Specific gravity (SSD) of coarse aggregate, S_{ca} = 2.69
- ✓ Specific gravity (SSD) of fine aggregate (sand), S_{fa} = 2.64

Procedure:

1. Target mean strength

   F_m = 33.5 N/mm^2

2. w/c ratio

- ✓ From Table 31 (SP: 23 - 1982),

  w/c ratio for compressive strength of 33.5 N/mm^2 = 0.50
- ✓ Maximum w/c ratio for given exposure condition = 0.50

  Thus, w/c ratio = 0.50

Concrete mix proportioning as per ACI Method

SP : 23 - 1982

TABLE 31 RELATIONSHIP BETWEEN WATER-CEMENT RATIO AND COMPRESSIVE STRENGTH OF CONCRETE

(*Clause* 6.1)

COMPRESSIVE STRENGTH AT 28 DAYS, kgf/cm^2	WATER-CEMENT RATIO, BY WEIGHT	
	Non-Air-Entrained Concrete	Air-Entrained Concrete
(1)	(2)	(3)
450	0.38	—
400	0.43	—
350	0.48	0.40
300	0.55	0.46
250	0.62	0.53
200	0.70	0.61
150	0.80	0.71

NOTE — Table 31 is from 'Recommended Practice for Selecting Proportions for Norm-' Weight Concrete' Reported by ACI Committee 211 (ACI Manual of Concrete Practice, Part I, 1979). American Concrete Institute, USA.

Concrete mix proportioning as per ACI Method

Requirements of ACI 318-89 for water/cement ratio and strength for special exposure conditions

Exposure condition	Maximum water/cement ratio, normal density aggregate concrete	Minimum design strength (MPa), low density aggregate concrete
Concrete intended to be watertight:		
a) Exposed to fresh water	0.50	25
b) Exposed to brackish or sea water	0.45	30
Concrete exposed to freezing and thawing in a moist condition:		
a) Kerbs, gutters, guardrails or thin sections	0.45	30
b) Other elements	0.50	25
c) In presence of de-icing chemicals	0.45	30
For corrosion protection of reinforced concrete exposed to de-icing salts, brackish water, sea water, or spray from these sources	0.40	33

Concrete mix proportioning as per ACI Method

3. Water content and cement content

For slump of 80 mm, 25 mm MSA, and non-air-entrained concrete,

Water content = W = 195 kg/m^3, Approximate amount of entrapped air = 1.5%

Cement content = C = 195/0.50 = 390 kg/m^3

SP : 23 - 1982

TABLE 32 APPROXIMATE MIXING WATER (kg/m^3 OF CONCRETE) REQUIREMENTS FOR DIFFERENT SLUMPS AND MAXIMUM SIZES OF AGGREGATES

(*Clause* 6.1)

SLUMP, cm	MAXIMUM SIZES OF AGGREGATES IN mm							
	10	12.5	20	25	40	50	70	150
	Non-Air-Entrained Concrete							
3 to 5	205	200	185	180	160	155	145	125
8 to 10	225	215	200	195	175	170	160	140
15 to 18	240	230	210	205	185	180	170	—
Approximate amount of entrained air in non-air-entrained concrete, percent	3.0	2.5	2.0	1.5	1.0	0.5	0.3	0.2
	Air-Entrained Concrete							
3 to 5	180	175	165	160	145	140	135	120
8 to 10	200	190	180	175	160	155	150	135
15 to 18	215	205	190	185	170	165	160	—
Recommended average total air content, percent	8.0	7.0	6.0	5.0	4.5	4.0	3.5	3.0

NOTE — Table 32 is from 'Recommended Practice for Selecting Proportions for Normal Weight Concrete' Reported by ACI Committee 211 (ACI Manual of Concrete Practice, Part I, 1979). American Concrete Institute, USA.

Concrete mix proportioning as per ACI Method

4. Coarse aggregate content

For 25 mm MSA, and sand with fineness modulus of 2.60,

Dry bulk volume of coarse aggregate per unit volume of concrete = 0.69

Coarse aggregate content = C_a

= 0.69 $\times$ Dry-rodded bulk density of coarse aggregate

= 0.69 $\times$ 1580 kg/m^3 = 1090.2 kg/m^3

SP : 23 - 1982

TABLE 33 VOLUME OF DRY-RODDED COARSE AGGREGATE PER UNIT VOLUME OF CONCRETE

(*Clause* 6.1)

MAXIMUM SIZE OF AGGREGATE mm	FINENESS MODULE OF SAND			
	2.40	2.60	2.80	3.00
(1)	(2)	(3)	(4)	(5)
10	0.50	0.48	0.46	0.44
12.5	0.59	0.57	0.55	0.53
20	0.66	0.64	0.62	0.60
25	0.71	0.69	0.67	0.65
40	0.76	0.74	0.72	0.70
50	0.78	0.76	0.74	0.72
70	0.81	0.79	0.77	0.75
150	0.87	0.85	0.83	0.81

Concrete mix proportioning as per ACI Method

First estimate of density (unit weight) of fresh concrete as given by ACI 211.1–91

Maximum size of aggregate (mm)	First estimate of density (unit weight) of fresh concrete (kg/m^3)	
	Non-air-entrained	Air-entrained
10	2285	2190
12.5	2315	2235
20	2355	2280
25	2375	2315
40	2420	2355
50	2445	2375
70	2465	2400
150	2505	2435

Concrete mix proportioning as per ACI Method

5. Density (unit weight) of fresh concrete

For maximum size of aggregate 25 mm,

First estimate of density of fresh concrete (from the table) = 2375 kg/m^3

6. Fine aggregate content

Fine aggregate content (F_a) = Density of fresh concrete – W – C – C_a

= 2375– 195 – 390 – 1090.2 = 699.8 kg/m^3

The volume method is more precise for calculating the required quantity of fine aggregate per unit volume of concrete and is given by the following equation:

$$V = \left[W + \frac{C}{S_c} + \frac{C_a}{S_{ca}} + \frac{F_a}{S_{fa}} \right] \times \frac{1}{1000} \qquad \text{................ (1)}$$

Concrete mix proportioning as per ACI Method

V = Volume of fresh concrete less volume of entrapped air

W = mass of water (kg) per 1 m^3 of concrete

C = mass of cement (kg) per 1 m^3 of concrete

C_a = mass of coarse aggregate (kg) per 1 m^3 of concrete

F_a = mass of fine aggregate (kg) per 1 m^3 of concrete

S_c = specific gravity of cement

S_{ca} = specific gravity (SSD) of coarse aggregate

S_{fa} = specific gravity (SSD) of fine aggregate

✓ Putting the values

V = 1 – 0.015 = 0.985 m^3, W = 195 kg/m^3, C = 390 kg/m^3, C_a = 1090.2 kg/m^3

S_c = 3.15, S_{ca} = 2.69, S_{fa} = 2.64 in Equation (1),

The value of F_a is given by;

F_a = 688.807 kg/m^3

Concrete mix proportioning as per ACI Method

7. Quantities

Water	Cement	Fine Aggregate	Coarse Aggregate
195 kg/m^3	390 kg/m^3	688.807 kg/m^3	1090.2 kg/m^3

Density of fresh concrete = 195 + 390 + 688.807 + 1090.2 = 2364.007 kg/m^3

Concrete mix proportioning as per British mix design method (DOE method)

- ✓ 28 day target mean compressive strength = 51.6 N/mm^2
- ✓ Type of exposure: Severe
- ✓ Nominal cover to steel reinforcement: 30 mm
- ✓ Maximum size of aggregate (MSA): 20 mm
- ✓ Aggregates of uncrushed type
- ✓ Workability: Slump = 55 mm
- ✓ Cement: OPC (ordinary Portland cement)
- ✓ Specific gravity of cement = 3.15

Concrete mix proportioning as per British mix design method (DOE method)

- ✓ Specific gravity (SSD) of coarse aggregate = 2.70
- ✓ Specific gravity (SSD) of fine aggregate (sand) = 2.70
- ✓ Fine aggregate: conforming to Grading zone 2

Procedure:

1. **Target mean strength**

   F_m = 51.6 N/mm^2

Concrete mix proportioning as per British mix design method (DOE method)

SP : 23 - 1982

TABLE 37 APPROXIMATE COMPRESSIVE STRENGTHS OF CONCRETE MIXES MADE WITH WATER-CEMENT RATIO OF 0.5

(*Clause* 6.3)

Type of Cement	Type of Coarse Aggregate	Compressive Strength (N/mm^2) Age (days) 3	7	28	91
(1)	(2)	(3)	(4)	(5)	(6)
Ordinary Portland Cement or Sulphate Resisting Portland Cement	Uncrushed	18	27	40	48
	Crushed	23	33	47	55
Rapid Hardening Portland Cement	Uncrushed	25	34	46	53
	Crushed	30	40	53	60

Concrete mix proportioning as per British mix design method (DOE method)

SP : 23 - 1982

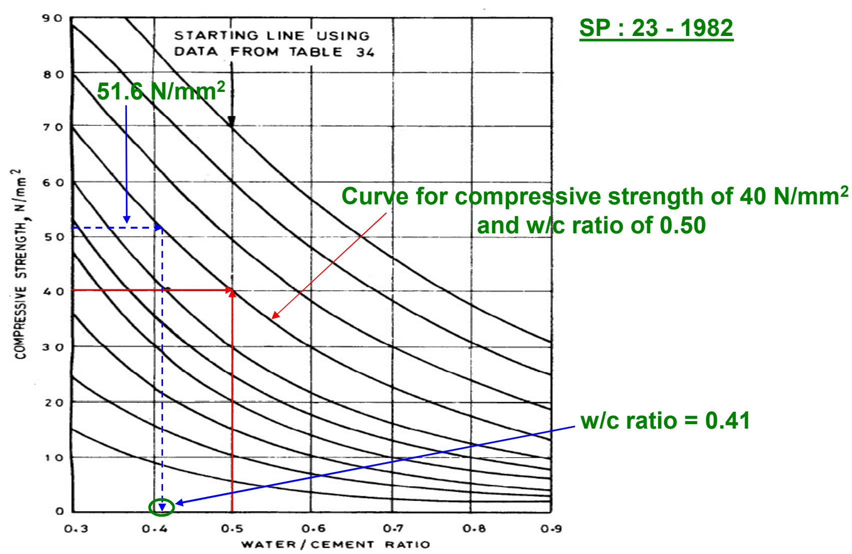


Fig. 43 Relationship Between Compressive Strength and Water-Cement Ratio

Concrete mix proportioning as per British mix design method (DOE method)

Requirements of BS 8110 : Part 1 : 1985 to ensure durability under specified exposure conditions of reinforced and prestressed concrete made with normal weight aggregate

Condition of exposure	Nominal cover of concrete (mm)				
Mild	25	20	20	20	20
Moderate	-	35	30	25	20
Severe	-	-	40	30	25
Very severe	-	-	50	40	30
Extreme	-	-	-	60	50
Maximum free water/cementitious material ratio	0.65	0.60	0.55	0.50	0.45
Minimum content of cementitious material (kg/m^3)	275	300	325	350	400
Minimum Grade (MPa)	30	35	40	45	50

This table is applicable when maximum size of aggregate is 20 mm.

Concrete mix proportioning as per British mix design method (DOE method)

SP : 23 - 1982

TABLE 38 APPROXIMATE WATER CONTENTS (kg/m^3) REQUIRED TO GIVE VARIOUS LEVELS OF WORKABILITY

(*Clause* 6.3)

	Slump (mm) →	0-10	10-30	30-60	60-180
	Vee-Bee (s) →	> 12	6-12	3-6	0-3
Maximum Size of Aggregate (mm)	Type of Aggregate				
(1)	(2)	(3)	(4)	(5)	(6)
10	Uncrushed	150	180	205	225
	Crushed	180	205	230	250
20	Uncrushed	135	160	180	195
	Crushed	170	190	210	225
40	Uncrushed	115	140	160	175
	Crushed	155	175	190	205

Note 1 — When coarse and fine aggregates of different types are used, the water content is estimated by the expression:

$$2/3\ W_f + 1/3\ W_c$$

where

W_f = water content appropriate to type of fine aggregate, and

W_c = water content appropriate to type of coarse aggregate.

2. Estimation of w/c ratio

- ✓ From Fig. 43, w/c ratio = 0.41
- ✓ As per BS 8110 : Part 1 : 1985, maximum free water/cementitious material ratio = 0.50
- ✓ Thus, w/c ratio = 0.41 (i.e., lower of the two values)

3. Water content and cement content

✓ For slump of 55 mm, 20 mm MSA, and uncrushed aggregates,

Water content = W = 180 kg/m^3

Cement content = C = 180/0.41 = 439.024 kg/m^3

Minimum cementitious material content as per BS 8110 : Part 1 : 1985 = 350 kg/m^3

Thus, C = 439.024 kg/m^3 (i.e., greater of the two values)

Concrete mix proportioning as per British mix design method (DOE method)

SP : 23 - 1982

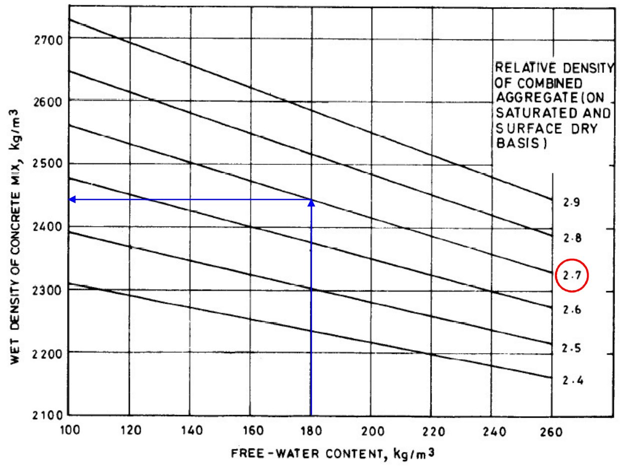


Fig. 44 Estimated Wet Density of Fully Compacted Concrete

Concrete mix proportioning as per British mix design method (DOE method)

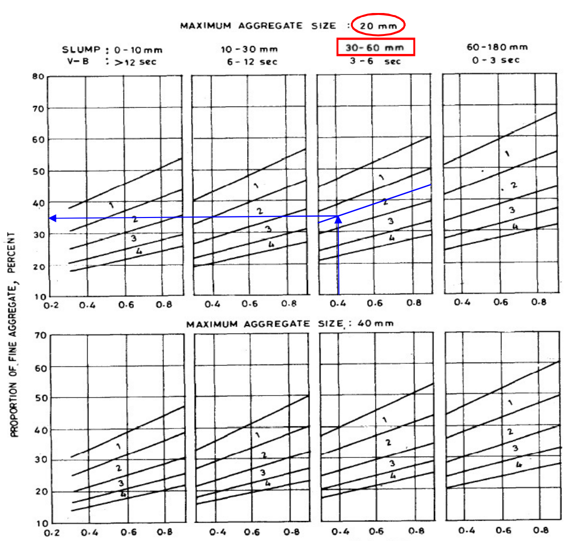


Fig. 45 Recommended Proportions of Fine Aggregate for Grading Zones 1, 2, 3 and 4

SP : 23 - 1982

Concrete mix proportioning as per British mix design method (DOE method)

4. Wet density of concrete = 2443 kg/m^3

5. Aggregate contents

Total aggregate content = 2443 – W – C = 2374 – 180 – 439.024 = 1823.976 kg/m^3

Proportion of fine aggregate = 34.8%

Fine aggregate content = 0.348 x 1823.976 = 634.744 kg/m^3

Coarse aggregate content = 1823.976 – 634.744 = 1189.232 kg/m^3

6. Quantities

Water	Cement	Fine Aggregate	Coarse Aggregate
180 kg/m^3	439.024 kg/m^3	634.744 kg/m^3	1189.232 kg/m^3

References

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